

Request for Assistance: Character Material Fingerprint Specification

Context

I have completed an N+3 normalization of my Material Universe — approximately 30 materials across 11 categories (Stone, Onyx, Pearl, Gemstone, Metal, Sheen, Organic, Fabric, Atmospheric, Card FX, FGE Signature). Each material exists as a clean, schema-correct object with consistent fields: `id`, `name`, `category`, `tags`, `accent`, `character`, `visual`, `prompt`.

This matrix is the palette layer of a larger system. It is not the deliverable. The deliverables are:

1. A premium NFT card series (Rookie Card format, 4 anchored characters, multi-tier editions)
2. Lookbooks generated as thematic traversals of the matrix
3. Greater scene renders composed from material interactions

The Specific Task

Before I build the Card Recipe Schema or the lookbook query engine, I need to produce the bridging artifact that makes both possible: **Character Material Fingerprints**.

I have four anchored characters:

- **Marcus Kane** — commanding Italian-heritage male, 193cm, salt-and-pepper hair, Armani pinstripe, authority register
- **Calista Voss** — 170cm, platinum-silver hair, green eyes, lean elegant control
- **Raven Voss** — 157cm, bronzed Latina complexion, Lean Elegant Muse Governor v1.0, sister/shadow dynamic with Calista
- **Isolde Voss** — Calista's shadow twin, obsidian black hair, Aurora green eyes, drop leader

For each character, I need a fingerprint composed of:

- **1 signature material** (the single matrix entry that *is* the character's visual identity)
- **3–5 supporting materials** (palette extensions that compose with the signature)
- **1 atmospheric material** (the environmental register the character moves through)
- **1 frame/edition material** (Card FX layer)

Each material in the fingerprint must reference a real `id` from the existing matrix.

Constraints

1. **Use only materials that exist in the matrix.** Do not invent new ones. If a character clearly needs a material that isn't in the matrix yet, flag it as a gap rather than filling it speculatively.
2. **Respect existing character-material bindings already encoded in the matrix.** Several materials have a `character` field that already names a character (e.g., `tahitian-pearl` → Raven Voss, `liquid-chrome` → Raven Voss Hot Metric). Honor those bindings; do not reassign.
3. **Sister-pairing logic for the Voss line.** Calista, Raven, and Isolde share a family visual grammar. Their fingerprints should overlap intentionally on at least one supporting material to encode kinship, while their signatures must be distinct enough to read as individuals at a glance.
4. **Marcus operates in a different register.** He should not share supporting materials with the Voss line except where the matrix forces it (e.g., shared atmospheric materials are acceptable; shared signature-tier materials are not).
5. **No prose embellishment in the output.** I want structured data I can drop directly into the next layer. Narrative reasoning belongs in a separate "rationale" field, not woven into the fingerprint itself.

Required Output Format

For each character, produce:

```
{
  "fingerprint_id": "[character_id].fingerprint.v[major].[minor]",
  "schema_version": "1.0",
  "matrix_version_dependency": "[matrix version this fingerprint was authored aga",
  "character": "[name]",
  "signature_material": "[matrix_id]",
  "supporting_materials": ["[id]", "[id]", "[id]"],
```

```

"atmospheric_material": "[matrix_id]",
"frame_material": "[matrix_id]",
"kinship_overlap": ["[id]"],    // for Voss sisters only; Marcus = null
"rationale": "[2-3 sentences explaining why these materials cohere as this char
"
gaps_flagged": ["[description of any missing material the matrix would benefit
"
changelog": "[for v1.0: 'Initial fingerprint.' For later versions: what change
}

```

Versioning Convention (must be applied)

Fingerprints are versioned independently from both the matrix and the card recipes. The receiver must apply the following rules:

1. Fingerprint ID format: `[character_id].fingerprint.v[major].[minor]`

- **Example:** `isolde.fingerprint.v1.0`
- Use the character's lowercase first name as `character_id` unless a different ID is already in use elsewhere in the system.

2. Major version bumps (v1.0 → v2.0) are required when:

- The signature material changes
- More than half the supporting materials change
- The character's visual identity is being intentionally rewritten

3. Minor version bumps (v1.0 → v1.1) are required when:

- A single supporting material is swapped, added, or removed
- The atmospheric or frame material changes
- The kinship overlap is adjusted
- A gap is filled by a new matrix entry that the fingerprint should now reference

4. `matrix_version_dependency` is mandatory. Every fingerprint must declare which version of the Material Universe matrix it was authored against. If the matrix is currently unversioned, the receiver should propose a starting version (suggested: `matrix.v1.0`) and apply it consistently across all four fingerprints.

5. `changelog` is mandatory even for v1.0. For initial versions, a single sentence is sufficient. For later versions, the changelog must state which fields changed and the reason.

6. **Backward compatibility:** When a fingerprint version bumps, any Card Recipe referencing the old version must continue to resolve until explicitly migrated. Do not assume cascading updates; the recipe layer will pin to specific fingerprint versions deliberately.
7. **No silent edits.** If the receiver believes a fingerprint should be revised after initial production, they must produce a new version with a changelog entry rather than editing the v1.0 in place.

Then produce a final **cross-character summary table** showing:

- Which materials are used by exactly one character (unique signatures)
- Which materials are shared across the Voss line (kinship layer)
- Which materials are unused so far (matrix coverage gap)

What I Will Do With This

Once the fingerprints are stable, I will use them to:

1. Specify the first 24-card season (4 characters × 6 cards each) by varying scene modifier and edition tier while holding the fingerprint constant
2. Generate the first lookbook as a single-character traversal (likely Isolde, since she's leading the Gumroad drop)
3. Build the Card Recipe Schema with the fingerprint as the inheritance root

What I'm Asking You For

Produce the four character fingerprints and the cross-character summary table, following the format and constraints above exactly. Flag gaps honestly. Do not optimize for completeness over correctness — if a character only has 3 strong supporting materials in the matrix and the rest would be reaching, give me 3 and note the gap.

If you need clarification on any character's existing matrix bindings before producing the fingerprint, ask before generating. Do not guess.